

LAB DIGITAL ASSIGNMENT 4

SUBJECT: NATURAL LANGUAGE PROCESSING LAB (PMDS606P)

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Course: M.Sc Data Science

```
In [10]: import csv  
import pandas as pd
```

Generating wordlist.csv of English vs German

```
In [11]: data = {  
    "English": ["water", "sun", "mother", "night", "book", "friend", "house", "bread", "love", "star"],  
    "German":  ["Wasser", "Sonne", "Mutter", "Nacht", "Buch", "Freund", "Haus", "Brot", "Liebe", "Stern"]  
}  
  
df = pd.DataFrame(data)
```

```
csv_path = "wordlist.csv"
df.to_csv(csv_path, index=False)
```

```
In [12]: df = pd.read_csv("wordlist.csv")
df.head()
```

```
Out[12]:
```

	English	German
0	water	Wasser
1	sun	Sonne
2	mother	Mutter
3	night	Nacht
4	book	Buch

Edit Distance (Levenshtein Distance) Method

```
In [13]: def edit_distance(word1, word2):
m, n = len(word1), len(word2)
dp = [[0] * (n + 1) for _ in range(m + 1)]

for i in range(m + 1):
    dp[i][0] = i
for j in range(n + 1):
    dp[0][j] = j

for i in range(1, m + 1):
    for j in range(1, n + 1):
        if word1[i - 1] == word2[j - 1]:
            cost = 0
        else:
            cost = 1
        dp[i][j] = min(
            dp[i - 1][j] + 1,
            dp[i][j - 1] + 1,
            dp[i - 1][j - 1] + cost
```

```
)  
return dp[m][n]
```

Output

```
In [17]: with open("wordlist.csv", "r") as file:  
        reader = csv.reader(file)  
        header = next(reader) # Skip header  
        for row in reader:  
            word1, word2 = row[0].strip(), row[1].strip()  
            distance = edit_distance(word1, word2)  
            if distance >= 3:  
                print(f"'{word1}' vs '{word2}' --> distance = {distance}")
```

```
'water' vs 'Wasser' --> distance = 3  
'sun' vs 'Sonne' --> distance = 4  
'mother' vs 'Mutter' --> distance = 3  
'night' vs 'Nacht' --> distance = 3  
'book' vs 'Buch' --> distance = 4  
'friend' vs 'Freund' --> distance = 3  
'house' vs 'Haus' --> distance = 3  
'bread' vs 'Brot' --> distance = 4  
'love' vs 'Liebe' --> distance = 4  
'star' vs 'Stern' --> distance = 3
```

```
In [ ]:
```